

EC1410-Fall 2026 Problem Set 7

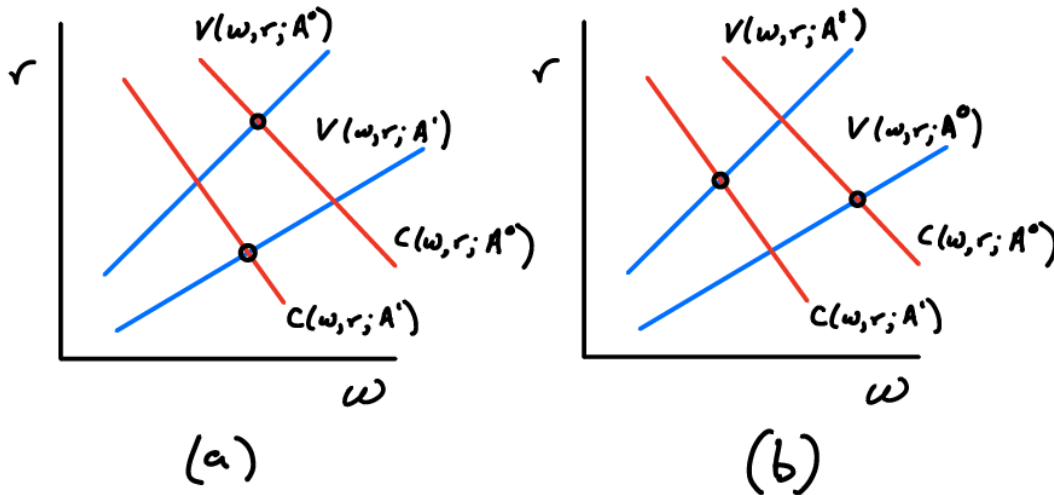
(Updated 7 August 2026)

Matt Turner

When you write up your answers, your goals should be to (1) be correct, and (2) convince your reader that your answer is correct. It is always helpful if your work is legible and if all steps are presented, possibly with a line of explanation. Answers that do not achieve these goals will not be awarded full credit. 100 points are possible. Points for each problem given in parentheses.

Problems

1. Consider the following two figures showing indifference curves and iso-cost curves for two levels of an amenity, where $A_1 > A_0$.



- (a) (10) For the economy in panel (a), does the amenity increase or decrease productivity? Does it increase or decrease utility and utility? Explain briefly.
 - (b) (10) For the economy in panel (b), does the amenity increase or decrease productivity? Does it increase or decrease utility? Explain briefly.
2. This problem asks you to solve for the households indirect utility function and describe indifference curves in (r, w) space. Let $U(c, l_c, A) = \bar{u}$ and assume the household problem is given by

$$\max_{c, l_c} U(c, l_c, A) = A c^{2/3} l_c^{1/3} \text{ such that } w = c + r l_c$$

- (a) (10) Solve the constraint for c . Plug your expression for c into the utility function.
 - (b) (10) Solve the maximum problem for l_c .

- (c) (10) Find the indirect utility function $V(w, r, A)$ by substituting demand for housing and consumption into $U(c, l_c, A)$.
 - (d) (10) Define an indifference curve by $V(w, r, A) = \bar{u}$. Solve for r in terms of $A, \bar{u}$, and w .
 - (e) (10) Evaluate $\frac{\partial r}{\partial w}$. What is the sign of this derivative?
Is A an amenity or dis-amenity from the perspective of the consumer? Explain briefly.
3. (30) This problem asks you to calculate the importance of amenity A in real terms. Assume you have data on rents, wages, and amenity A for a cross-section of cities. That is, your data is $\{r_i, w_i, A_i\}$ for a set of cities $i = 1, \dots, J$. You may also assume that housing expenditure is one-third of the city wage. Describe the regressions you would run, and any subsequent analysis you would do, to determine the importance of amenity A in real terms (that is, as a share of the city wage).